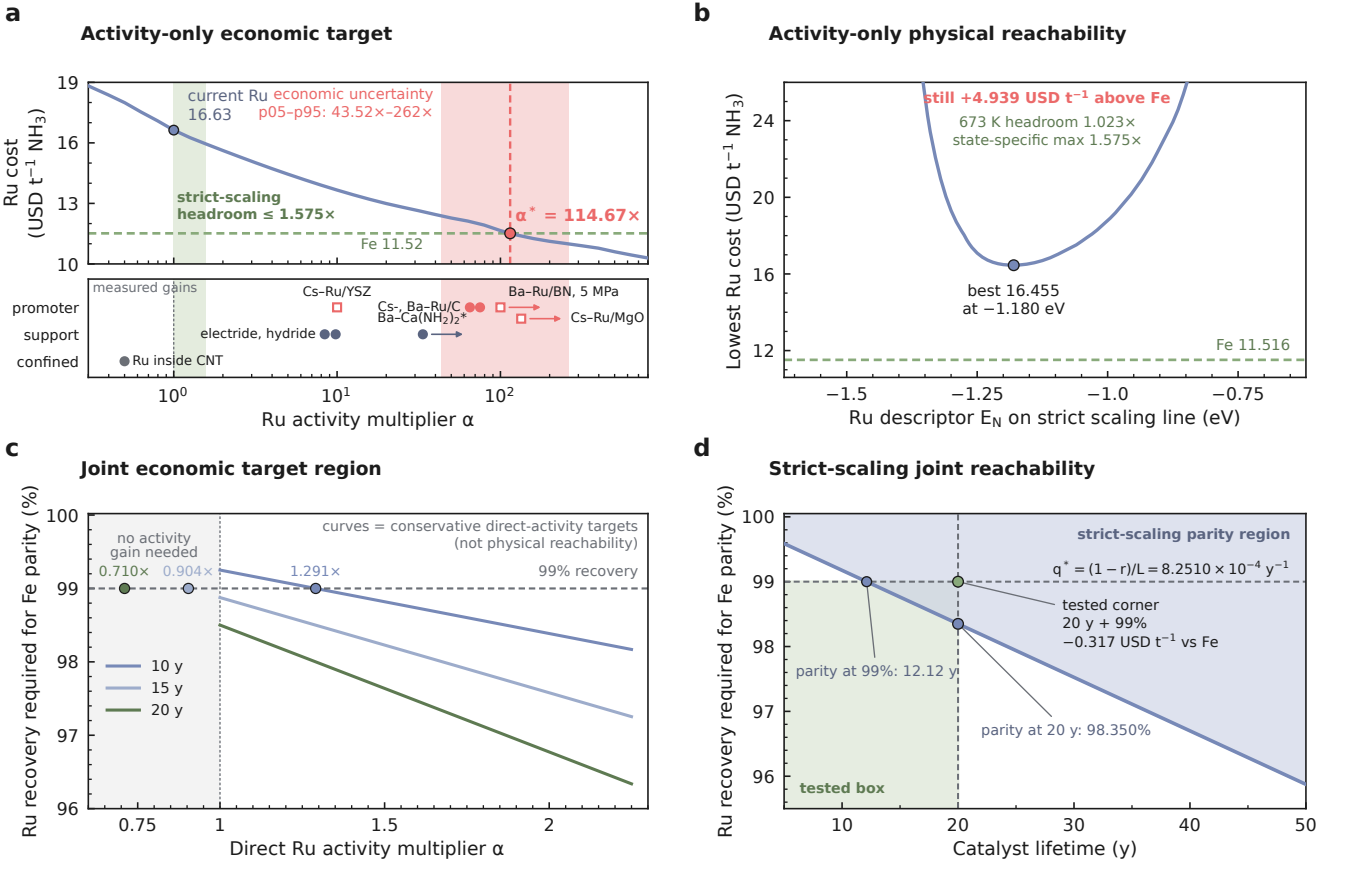
a
Activity-only economic target
Ru cost (USD t⁻¹ NH₃)
current Ru 16.63
economic uncertainty p05–p95: 43.52×–262×
strict-scaling headroom ≤ 1.575×
α* = 114.67×
Fe 11.52
measured gains
promoter
support
confined
Cs–Ru/YSZ
Ba–Ru/BN, 5 MPa
Cs-, Ba–Ru/C
Ba–Ca(NH2)2*
Cs–Ru/MgO
electride, hydride
Ru inside CNT
Ru activity multiplier α
b
Activity-only physical reachability
Lowest Ru cost (USD t⁻¹ NH₃)
still +4.939 USD t⁻¹ above Fe
673 K headroom 1.023×
state-specific max 1.575×
best 16.455
at −1.180 eV
Fe 11.516
Ru descriptor E_N on strict scaling line (eV)
c
Joint economic target region
Ru recovery required for Fe parity (%)
no activity gain needed
0.710×
0.904×
1.291×
curves = conservative direct-activity targets
(not physical reachability)
99% recovery
10 y
15 y
20 y
Direct Ru activity multiplier α
d
Strict-scaling joint reachability
Ru recovery required for Fe parity (%)
strict-scaling parity region
q* = (1 − r)/L = 8.2510 × 10⁻⁴ y⁻¹
tested corner
20 y + 99%
−0.317 USD t⁻¹ vs Fe
parity at 99%: 12.12 y
parity at 20 y: 98.350%
tested box
Catalyst lifetime (y)